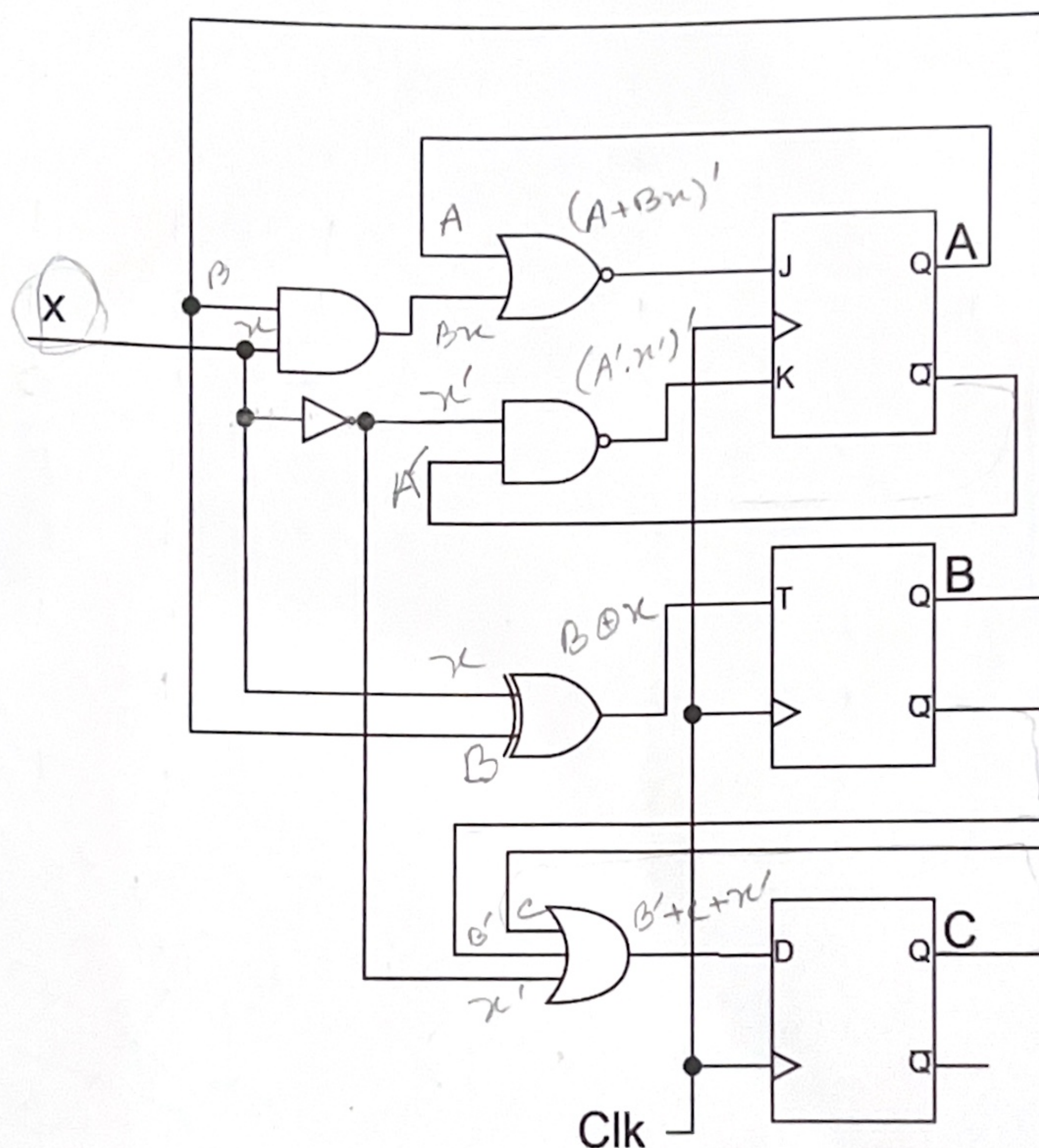


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CSE260 Quiz 4

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Q1. Design the state diagram for the following circuit diagram. Show all necessary and relevant tables and equations.



Present states: A, B, C Next states: A⁺, B⁺, C⁺

Inputs: JA, KA, TB, DC, X

Outputs: None (external)

eqns $JA = (A + Bx)' = A' \cdot (Bx)' = A' \cdot (B' + x')$

$KA = (A'x')' = A'' + x'' = A + x$

$TB = B \oplus x$ $DC = B' + C + x'$

State Diagram

x	A	B	C	JA	KA	TB	DC	A^+	B^+	C^+
→ 0	0	0	0	1	0	0	1	1 ✓	0 ✓	1 ✓
→ 0	0	0	1	1	0	0	1	1 ✓	0 ✓	1 ✓
→ 0	0	1	0	1	0	1	1	1	0	1
→ 0	0	1	1	1	0	1	1	1	0	1
→ 0	1	0	0	0	1	0	1	0	0	1
→ 0	1	0	1	0	1	0	1	0	0	1
0	1	1	0	0	1	1	1	0	0	1
0	1	1	1	0	1	1	1	0	0	1
→ 1	0	0	0	1	1	1	1	1 ✓	1 ✓	1 ✓
→ 1	0	0	1	1	1	1	1	1 ✓	1 ✓	1 ✓
→ 1	0	1	0	0	1	0	0	0	1	0
→ 1	0	1	1	0	1	0	1	0	1	1
→ 1	1	0	0	0	1	1	1	0	1	1
→ 1	1	0	1	0	1	1	1	0	1	1
1	1	1	0	0	1	0	0	0	1	0
1	1	1	1	0	1	0	1	0 ✓	1 ✓	1 ✓

T	B^+	TB
0	0	0
0	1	1
1	0	1
1	1	0

0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

State Diagram

